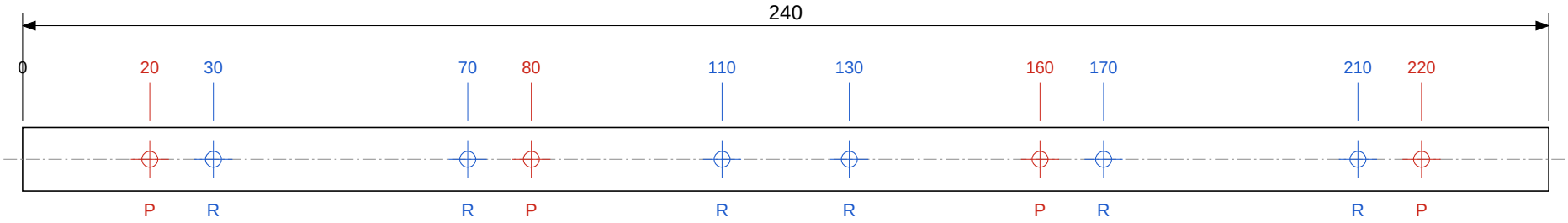
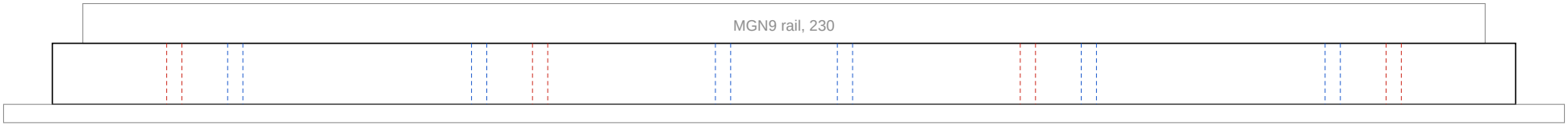
iPhone-Ian: RAIL RISER BAR (qty 2, identical)

1:1 drilling template. Print at 100% / Actual size and check the scale bar before punching.

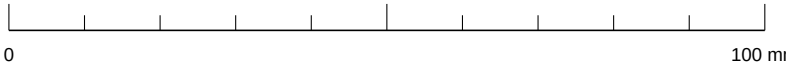


TOP VIEW, 1:1. Cut out along the outline, tape it to the bar, centre-punch each crosshair.



SIDE VIEW, 1:1. Every hole goes straight through. R screws come down through the rail, P screws come up through the plate.

base plate, 3 mm



Scale check: this must measure exactly 100 mm. If not, reprint at Actual size.

R rail screws, 6 holes: 30, 70, 110, 130, 170, 210

P plate screws, 4 holes: 20, 80, 160, 220

mm from either end

NOTES

1. Stock: 6061 aluminium, 10 x 10 x 304.8 mm (12 in). Cut each to 240 mm, square the ends, deburr.
2. All holes on the centreline, 5 mm from each side. Centre punch, drill Ø2.5 straight through, tap M3 x 0.5 all the way through.
3. Cutting oil on the drill and tap. Back the tap off every half turn to break the chip.
4. The pattern is symmetric: measure from either end, drill from either face. Both bars are identical.
5. Before drilling the R holes, lay the rail 5 mm in from each end and check its holes sit at 10, 30, 50 ... (20 mm pitch).
6. P holes must match the plate within 0.2 mm along the bar. Best: mark them through the plate's slots once it arrives.
7. Screws: M3 x 8 in every hole (rail from above, plate button heads from below).